RYAN CHEN RUI YANG

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Education

National University of Singapore Aug 2023 - Expected May 2027

Bachelor of Computing (Hons.) Computer Science, GPA 4.45/5.00

- Modules taken: Operating Systems, Software Engineering, Data Structures and Algorithms, AI & Machine Learning
- **Distinction Certificate** in Networks and Distributed Systems, AWS Certified Solutions Architect Associate
- AY24/25 NUS Raffles Hall Videography Club, Vice Chairperson

Work Experience

CSIT, Software Engineering Intern, Singapore Oct 2026 - Present

- Cloud native application development of full-stack systems.

DBS Bank Ltd, Cloud Infrastructure & Automation Intern, Singapore May 2026 - Oct 2026

- Designed and deployed cloud monitoring platforms with compliance validation, leveraging Playwright browser automation and API-based pipelines to eliminate 80% of manual workflows.
- Integrated checksum validations and audit trails as a critical regulatory layer to detect document modifications.
- Automated routine infrastructure operations for reports, logging and scripting via parallel, asynchronous subtasks to improve operational efficiency.

NUS, SWE Undergraduate Teaching Assistant, Singapore Aug 2025 - Present

- Mentored students in Java, explaining Git, CI/CD pipelines, SDLC, software testing, and software architecture for scalable and efficient systems, achieving a 100% pass rate with B+ or better scores, and a 4.5/5.0 tutor rating.
- Reviewed and graded student code against established coding principles, providing timely feedback to reduce errors in future submissions by about 70%.

HTX, Cloud & Software Engineer Intern, Singapore May 2025 - Aug 2025

- Implemented a REST API framework in Microsoft Power Automate to automate image ingestion pipeline for HTX HR, reducing manual data entry by 80% every week, and ensuring 100% availability for continued usage.
- Automated tracking of external contractor leave, connecting a SharePoint data source with an external interface, eliminating 20 hours spent monthly on data syncing and email tracking across departments.

Projects

CRAG, Course-Grounded AI Learning Platform, NUS Final Year Project May 2026 - Present

- Designed and built a full-stack AI learning platform using React, FastAPI and PostgreSQL/pgvector, combining corrective retrieval over course materials with source citations, instructor-reviewed answers, customised learning modes, quizzes and learning analytics across web and Discord interfaces.
- First-authored a research paper accepted for IEEE TALE 2026, linking a 42-student needs assessment to course-grounded AI design and a proposed framework for evaluating reflective learning.

EcoTrack, Team Co-lead, NUS LifeHack 2025 Hackathon Winner Jun 2025 - Jul 2025

- Built an AI-driven "Wikipedia of product lifecycles" integrating a Chrome extension, interactive dashboard, and Telegram bot to generate community-verified sustainability scores and lifecycle summaries within a 48-hour timeframe.
- Engineered backend pipeline with OpenAI embeddings, Postgres (vector storage), and a custom tree-based algorithm to scale lifecycle summarisation, maintain performance with sub-500ms latency, and handle 1000+ events per product.

TsFraudPmo, Team Co-lead, DSTA BrainHack Hackathon Finalist May 2025 - Jun 2025

- Full-stack development of a real-time scam detection platform, integrating a React Native mobile app and Telegram bot to enable instant user reporting and rapid dissemination of verified scam intelligence.
- Implemented custom LLM pipelines, utilising prompt engineering, semantic embedding, and real-time input processing, transforming raw user reports into over 100 validated insights and concise summaries.

Technical Skills

Languages: Python, TypeScript, JavaScript, Java, SQL, C++, C#, C, HTML, CSS
Web & Backend: React, React Native, FastAPI, Django, Angular, REST APIs
AI & Retrieval: RAG, LangChain, LangGraph, LLM integration, prompt engineering, vector embeddings
Databases: PostgreSQL, pgvector, Supabase, MariaDB
Testing & DevOps: Git, GitHub Actions, Pytest, Playwright, Gradle, Codecov
Cloud & Automation: AWS, Azure, GCP Vertex AI, Vercel, Microsoft Power Platform